

Intrusion Detection System to detect impersonation attacks in IoT networks

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Abstract— The instant and rapid growth of interrelated communication and computing devices in the Internet which electronically binds the various systems in social and physical way and the raising levels in quantity of sensitive and significant data conveyed and transferred has posed a threat to concerns regarding the process of securing the privacy, safety in networks of devices of IoT. Upshot of the high count of devices being connected, they might be prone to cyber-attacks by illicit users. So there is always a need and demand for an excellent Intrusion Detection System (IDS) which identifies and prevents unauthorized access into the network system involving IoT devices. This proposed work presents Support Vector Machine (SVM) which is deployed and captures the network attributes from the given dataset for recognizing impersonation attack. We have utilized Aegean Wi-Fi Intrusion Dataset- (AWID), and experimental findings demonstrate that the performance and interpretation of the model proposed concerning to the detection rate, accuracy, and F1-score while identifying the impersonation attacks.

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I. INTRODUCTION

As a broadening network interconnecting several kinds of devices in IoT network and inclined towards the most commonly accepted in the domain of health and care, merchandises, production and transporting markets and in industrial applications [1]. The applications of IoT communications networks as depicted in Figure 1. Increasing security and protective mindset of different individuals, society, and the nations globally in view to the expanding portable electronic devices. The IoT network contains gadgets used in every single day which have the different levels of capacity to be intelligent and connected being either sophisticated computers and mobile devices. The amalgamation or integration of divergent technologies makes the security as a more significant aspect. The attacks which disturb the operations of banking and healthcare sector leads to huge loss in both time and money for concerned authorities. This might result in worldwide emergencies. It encompasses a network of interconnected objects equipped with sensors, software, and control systems. Technological progress, particularly in machine learning, has played a

pivotal role in shaping the evolution of IoT technology. The influence of IoT applications is progressively permeating diverse domains, establishing a ubiquitous presence in nearly every facet. Security is essential part in IoT networks because communication devices and data are vulnerable to cyber threats. The IoT is very much practical today as it is utilized widely to furnish the online transactions in e-commerce to make the activities of accessing and billing at various industrial sectors. Thus, businesses must be aware of the novel kinds of IoT cyber security threats to safeguard the devices and themselves as the data simultaneously being collected. The primary way in which attacks can be started is through devices. Memory, web interface, network services, firmware, and physical interface are the few parts of a device where threats can come from [2]. Impersonation attack, Physical attack, Encryption attack, DoS (Denial of service attack), Firmware hijacking, Man-in-the-middle attack, Ransomware, Eavesdropping, Privilege escalation, Brute force password attack are a few attacks where hackers are attempting to get down the data from the devices by using these techniques. Data and networking devices in networks of different IoT gadgets are susceptible to various cyber-attacks. The list of various cyber-attacks are emphasized in Figure 2.

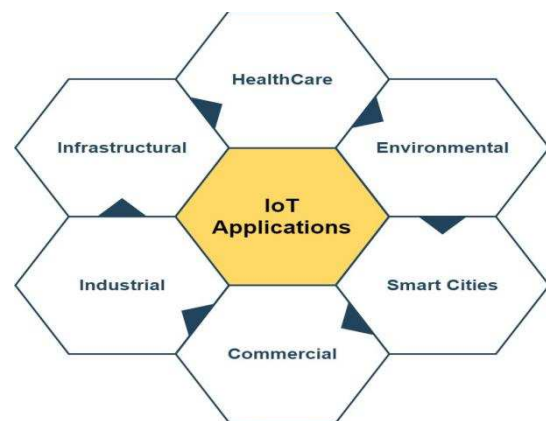


Fig 1: Applications of IoT Communications Network

1) Denial-of-Service (DoS) attack: A DoS attack is described as an intentional and malicious effort to disturb the regular flow of data packets and traffic to a particular or specifically targeted server, service, or network [3]. Due to various resource constraints in networks of IoT gadgets, DoS

attacks disrupt the services in network like data loss corruption and compromised security.

2) **Replay Attack:** It is a routing security threat wherein an intruder records data from other systems and subsequently transmits this data to neighboring nodes. This kind of attack leads to nodes altering their routing information with malicious data, thereby generating an incorrect network topology [4].

3) **Man-in-the-Middle attack:** This attack occurs when a hacker infiltrates the communication and transactions of data packets between two distinct systems. By covertly intercepting the exchange of information between two parties, this form of attack deceives the recipient into believing they are receiving a genuine message [5].

4) **Black hole attack:** A malicious user selectively discards each application packet received from child nodes in the directed graph without relaying on to the IoT networks[6]. In this attack, malicious users can exploit various vulnerabilities in routing protocol to compromise authorized nodes and to selectively drop packets from the communication path.

5) **SQL Injection attack:** An attacker seeking to perform SQL injection manipulates a conventional SQL query to exploit vulnerabilities arising from non-validated input in a database[7].

6) **User-to-Root (U2R) attacks** represent a kind of intrusion where the unauthorized user endeavour to acquire the network resources by posing as a regular user[8].

7) **Sinkhole attack:** While pertaining to sinkhole attack, a unauthorized device has the potential to alter the network structure by manipulating rank values. The malicious node, in turn, can selectively forward or discard all network data. This involves the compromise of a node, enabling an intruder to divert traffic and obstruct the information reaching to the destination node [9].

8) **Phishing attacks:** Phishing constitutes a mere class form of cybersecurity attack wherein malicious individuals send deceptive messages, assuming the name and identity of a trusted person [10].

9) **Sybil attack:** A Sybil attack involves an assailant assuming numerous identities simultaneously, posing a significant challenge when interfacing with a P2P network. This sort of attack manipulates the network by establishing multiple fraudulent identities, enabling the attacker to be dominant and control over the entire network [11].

10) **Cloning attacks :** In a clone node attack, a malicious actor seeks to physically capture the confidential credentials of deployed devices in network of IoT domain in intention to create clones that closely resemble the original devices [12]. Impersonation attack [13] is a kind of targeted phishing attack where a malicious person pretends to be someone else or other entities to steal sensitive data from unsuspecting employees using social engineering tactics. Hackers attempt to trick the victim into transferring money, giving up sensitive information, or providing business login credentials to leverage cyberattacks and gain illicit and unauthorized or illegal access to device systems and networking systems.

The above-mentioned are the common representations of other different kinds of attacks which fall under the same

group as these attacks. Some attacks may trick the system to fetch the information or to inject the malwares or viruses. But some attacks try to use of the vulnerabilities on the side of the user entities and force the authorized users to share or leak the data directly from their side. These kind of attacks are exercised to retrieve the critical data of the organizations.

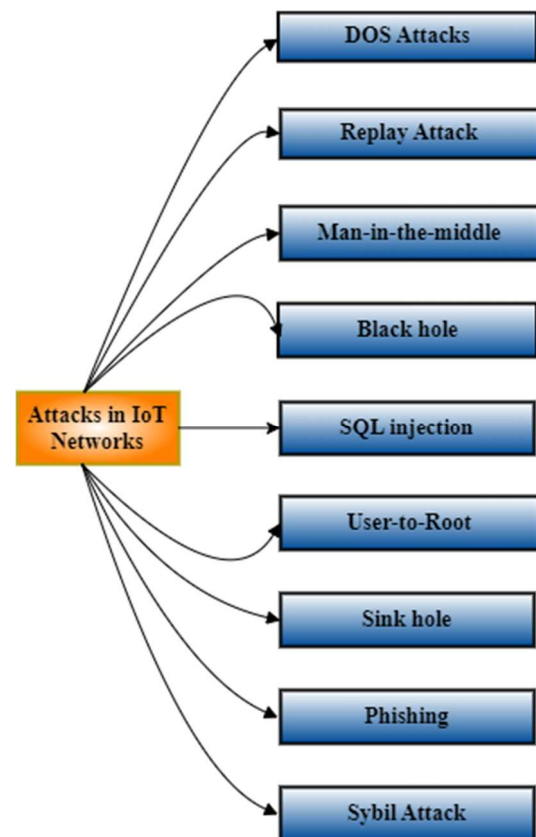


Fig 2: Cyber-attacks in IoT communication networks

Impersonation Attacks can take place by making use of resemblance to the original user identity, say E-mail IDs. E-mail IDs with a minute difference from the legitimate user are utilized in this form of attack, making this other form of impersonation attack differ from the phishing cyber mechanism. This sort of attack can be prevented when the user is adapted with some security measures. Proper security, double-checking sensitive information, Ensuring organizational level security, Protection of User Identity are the sensitive security measures required.

This paper is organised as follows. Section II introduces literature survey, section III depicts methodology and section IV depicts Conclusion and Future work.

II. LITERATURE SURVEY

The section of Literature Survey provides quick review of the status of various intrusion recognition techniques used in IoT communication networks to identify cyber threats.

Aminanto *et al.* [14] proposed the D-FES (Deep Feature Extraction and Selection) method, which utilizes stacked techniques of feature extraction and weighted feature selection to detect the attacks of the form of impersonation in Wi-Fi networks. AWID dataset was well referred for the Wi-Fi network benchmark dataset. Accuracy, Detection Rate,

Precision, False Alarm Rate, F1 score are utilized to estimate the proposed techniques. The shortcomings and disadvantages of this experiment is, it is emphasized only impersonation attack but not others and use inadequate computing power memory, and power supply.

Firdaus *et al.* [15] introduced prime features and ideal attributes utilizing a genetic search-based technique and static analysis for detecting the malware in android applications. They adopted Genetic search(GS) based method and used Benign and Malware datasets to detect malware activities in android applications. For future study, they recommend adding benign objects or data to the dataset to improve the detection results, for instance, to gain a lower FPR and increase the TPR value.

Parker *et al.* [16] developed deep learning model to detect cyber-attacks in IoT networks. CLS-AWID was utilized in this proposed model to detect IoT intrusion. detection rate, precision, accuracy, false alarm rate are the evaluation metrics used to judge the proposed technique. With regard to future work, they intend to explore the capability achieved by the proposed model to detect other attacks those are distinctly possible to occur within an IoT network and demonstrate the lightweight essence of the algorithm by deploying it within a resource constrained device.

Azar *et al.*, [17] proposed intrusion detection or diagnosis algorithm based on three machine learning classifiers. This paper provides three various methods which are: Firstly, Information Gain. Secondly, Gain Ratio. Thirdly, Correlation Feature Selection. These security convention and practices are exerted for selecting and ranking features then select combine the best top ranking features. The adopted model is deployed on KDD dataset to review the classification performance [18]. Time consuming for training classifier is the limitation of the put forth technique in this paper.

Peiying *et al.*, [19] proposed an alarm intrusion detection algorithm (FWP-SVM-GA) rooted on the Genetic Algorithm (GA) and SVM algorithm suitable for a human centered smart IDS. In compliance with the optimal feature subset, the feature weights and parameters of SVM are simultaneously optimized. The datasets are sourced from KDD.

Various obstacles and difficulties in developing an IDS to identify cyberattacks within IoT networks. Here under are the major of the issues and challenged encountered in designing process:

- IoT terminals are diverse and integrated which make IoT networks more prone to various intrusion attacks.
- In IoT networks a great number of interaction between more number of devices, which creates the dimension of network data larger. It is arduous to classify and take measures against various classes of attacks from massive data.
- Data mining statistics-based models and other algorithms have proven to furnish and satisfy the protection and security. However these methods are onerous and exhausting when it comes to adapting to fast-developing attacks.

- Intrusion prevention tools and signature based intrusion capturing and identifying methods cannot be effective against modified attacks.

The major and key contributions of this presented work are such as follows:

- 1) The intrusion detection model we have put forth is built on paradigm of expert machine learning functions to identify cyber threats.
- 2) In this work we have implemented Support Vector Machine (SVM) on AWID data set to detect impersonation attack.
- 3) The experimental findings are compared by assessing the worth of the outcomes from proposed SVM abled intrusion detection model.

III. METHODOLOGY

Under the section of Methodology, we set forth the intrusion recognition and detection system developed and presented, which includes three steps: data collection, data preprocessing, and model building. First, we do feature extraction on AWID dataset and then input the processed data for the model training and utilize it for detection of impersonation attack in the testing phase. The process flow of the propounded model is showcased in Figure 3.

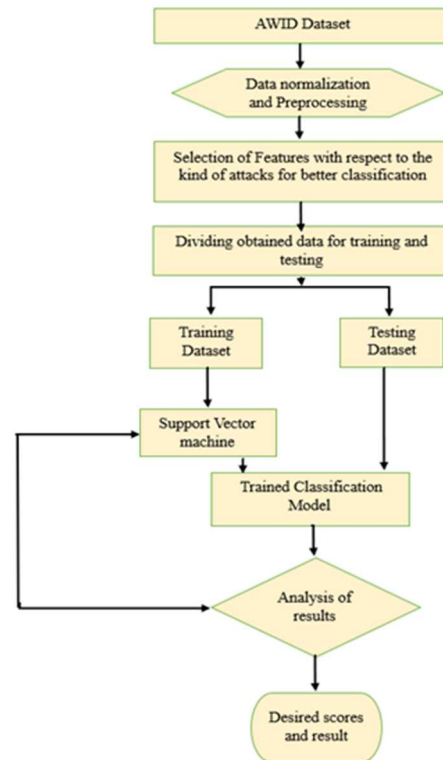


Fig 3: System flow of the presented SVM-based Intrusion Detection Model

A. Data Source

In this proposed system, we have utilized AWID dataset to evaluate our model. The above model is developed on the expert systems of learning tools using Python in the Jupyter Notebook. AWID is the open one and most widely utilized intrusion detection dataset which is AWIDAs it is distinctive by characterized data and unbalance between attack and normal data, which might affect the evaluation of the accomplishment and performa by the Intrusion IDS. The

dataset consists of fifteen classes and four classes respectively. The four contrast classes are classified that boost attacks belong to, including impersonation, flooding, and injection, and the class normal, while the other redundant dataset consists of the names of the initiated attacks and the normal class. The count of training records of each lower dataset is 1,795,575, and the count of test samples is 575,643. The count of features add up to 156, recognizing the WLAN fields along with physical layer metadata.

B. Data Preprocessing

As collected data should be preprocessed to remove noise, perform data normalization, and feature extraction. This step is crucial in bringing down the facets of the dimensions of the data, making it easier to analyze. The procedure of preferring and selecting of relevant features by the algorithm is critical to make decisions if the packet flow has the normal or malicious traffic. This step helps to reduce false positives and false negatives. Feature extraction is the important process for decreasing the considered feature variables by obtaining most important features from the dataset.

C. Model Training for classification

This propounded system was implemented and realized through SVM (Support Vector Machine) while utilizing the python machine learning packages and libraries which resulted trained binary classifier for impersonation attack in network topology of IOT gadgets. SVM being a one class classifier and a two-class classifier giving the accurate results and even tracking the outliers makes it the most suitable technique interpret and alert about the different attacks in and outside the network. SVMs are recognized as popularly utilized in numerous applications like text categorization, image classification, and bioinformatics. Their effectiveness, especially in high-dimensional spaces, and their propensity to steer on the complex datasets make them a popular choice in machine learning.

As the growing number of unknown kind of attacks effect in the raising complexity of non-linear problems, posing a challenge for most accepted techniques of machine learning and erudition in the operation of determining attacks by intelligent intrusion recognition and prevention systems, SVM diminishes the variable complexity by mapping the multidimensional and labelled data with large count of features into a high-dimensional space. The SVM Generated hyperplane(s) being member of affine family of less dimensional multi subspaces which are very alike to linear spaces makes the task of classification simple. Hence producing the divisions within data space optimally by increasing the margin or decision boundary of hyperplanes among datapoints belonging to different data classes as conceptualized by Figure 4. Major data entities or points concentrated around the margin or hyperplane are termed as Support Vectors, which optimize the position of hyper plane by computing the distances of individual class's data points to the hyper plane to take better classification decisions. Figure 4 above denotes the hyper plane obtained in a simple two class classification describing the support vectors, margin between support vectors and hyper plane separating the data points which either belong to legitimate or attacked class.

The proposed intrusion detection framework consists of two main processes, namely phases of testing and training. During the training stage, a dataset comprises of labeled samples will be obtained. Features selected from the network dataset form the initial stages in the implementation of algorithm. The selected labeled features are then utilized in training duration of the system for classification. Then, to check the throughput of the training model, similar features are made utilized from the phase of training are selected from the test samples. The unlabeled test samples will be given to the trained classifier for classifying the normal networking data records or attacked data records .

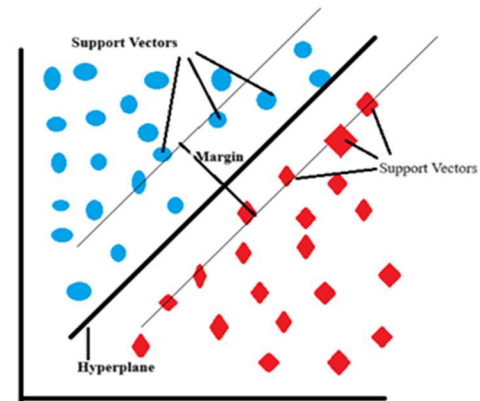


Fig 4: Procedure of SVM

D. Experimental Results

The proposed intrusion recognition model is implemented using Python in the Jupyter Notebook. Confusion matrix being the most utilized criterion to realize and analyze the machine learning model performance, especially for classifying the two classes and making the decision accordingly, which is the scenario of this study.

With the benefit from the metrics of confusion matrix, the computed evaluation measures below are utilized to measure the proficiency of the IDS model presented here. The evaluation metrics utilized in this gauging mechanism are accuracy, Detection Rate (DR), F1-measure (F1), False Alarm Rate (FAR). The six indicators listed below are used to judge the IDS in accordance with the definition above:

Accuracy: The correlation and ratio of the predicted outcome's accuracy to the entire sample count, which may be put in to measure the model's overall accuracy in classifying samples [20-22].

F1-Score: The merit of Harmonic mean between recall and precision is referred to as F1 Score.

Detection Rate: The right percentage of all positive examples serves as a proxy for coverage, which validates how well the classifier can identify positive occurrences.

False Alarm Rate (FAR): FAR, which accounts for a part of all real negatives, is projected to be positive but is in fact negative [23].

Table 1 demonstrates the contrast and comparison findings of the propounded model in terms of detection rate, F1 score, accuracy and false alarm rate. Experimental results evidently show that, the propounded system performs well in detecting impersonation attack in topology of IoT devices of a network in contrast with other existing and diverse models. Figure 5 denotes comparison between the results of the proposed impersonation attack model against the pre-existing models in accordance to detection rate, accuracy and F1-Score.

Table 1: Collation and Comparison of findings of the presented IDS

Method	Accuracy	DR	F1 Score	False Alarm Rate
Proposed method	99.98	100	100	-
Lee et al. [24]	98.22	97.64	-	1.20
Rahman et al. [25]	97.60	99.91	99.95	0
Cao et al. [26]	99.97	99.09	98.51	0.17
Bharathi et al. [27]	98.23	97.65	-	1.21

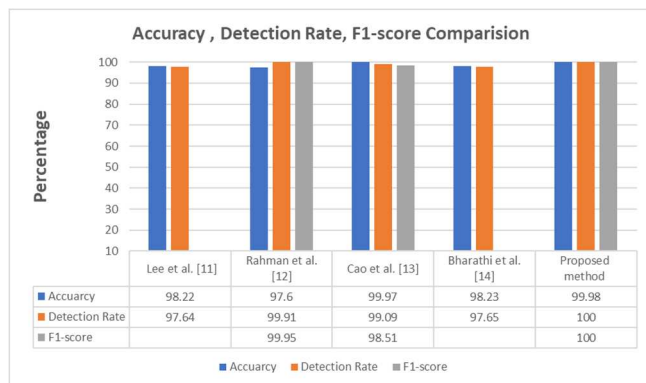


Fig 5: Comparison of outcomes of the presented model with existing models

IV. CONCLUSIONS

This paper introduces the creation of an expert system through the machine learning-driven method for an IDS that satisfies and enhances the capability to be deployed and executed directly on devices with limited resources. As there is raise in number devices connected and communicated through an internet. Devices and information in IoT communications network are prone to various cyber-attacks. However, resource limitations are inherent in IoT system or network including restricted energy resources, computational power, and memory, render conventional space ML-based IDS unsuitable for deployment on such devices. This study reviews and aims to implement a lightweight ML-based IDS adapted for resource-constrained IoT devices. It is very necessary to structure an intrusion identification technique in network topology of IoT system of connected devices to recognize attacks. In this article, an intrusion diagnosis and detection technique was designed on the foundation machine learning methods. We have utilized AWID dataset to valuate the proposed intrusion identification and detection model on the basis of SVM. We have used SVM for selecting the features and capturing the impersonation attack. The model is preferred to be exhaustively trained and tested on

additional datasets with their own features existing within the IDS research to evince its usefulness and effectiveness. SVM was the ML technique used to distinguish the normal and attacked data. SVM operates as a classifier and it classifies the dataset into two classes. Therefore the results show that the SVM classifier divides the dataset into binary classes i.e the normal data and the data obtained from previous attacks. The presented model performs well in detecting impersonation attacks as comparable with the state-of-the-art methods. However, in the coming future, the proposed model is to be extended to detect other cyber-attacks on IoT networks.

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